MODERN PHYSICS ORAL EXAM QUESTIONS [PRELIMINARY!]

Experimental basis of quantum mechanics

1. **Discovery of the electron** Cathode ray tubes, Millikan experiment
2. **Quantization of light** Line spectra, blackbody radiation, Planck's radiation law
3. **Photoelectric effect** Stopping voltage, work function
4. **X-rays** Bremsstrahlung, Bragg diffraction on crystals
5. **Rutherford model of the atom** Thompson's model, Rutherford scattering experiment

Quantum mechanics

6. **De Broglie waves** De Broglie wavelength, Bohr's quantization condition, electron scattering
7. **Schrödinger equation** Wavefunction, time-dependent, and time-independent equation
8. **Expectation values** Probabilistic interpretation of the wavefunction, expectation values
9. **Simple quantum-mechanical systems** Particle in infinite potential well, particle in harmonic potential, tunneling
10. **Hydrogen atom** Solutions of the Schrödinger equation for Coulomb potential, quantum numbers
11. **Angular momentum and spin** Angular momentum operators, Stern-Gerlach experiment, spin operators
12. **Electronic structure of the elements** Periodic table, total angular momentum, Hund's rules

Special relativity

13. **Michelson–Morley Experiment** Disproving the existence of ether
14. **Postulates of special relativity** Einstein's postulates, Lorentz transformation, invariant spacetime interval
15. **Relativistic effects** Length contraction, time dilation, muon lifetime, relativistic Doppler effect
16. **Relativistic dynamics** Relativistic energy, momentum, four-momentum, mass-energy equivalence
17. **Compton scattering**

Nuclear and Particle Physics

18. **Structure of the nucleus** Discovery of the neutron, nuclear density, nuclear forces
19. **Radioactive decay** Alpha, beta and gamma decay, decay chains, radioactive dating
20. **Nuclear stability** Empirical energy formula, stable isotopes, fission, fusion
21. **Relativistic quantum mechanics** Dirac equation, interpretation of the four-component wavefunction, antiparticles, Feynman diagrams
22. **The standard model** Leptons, neutrinos, quarks, hadrons, mesons, fundamental forces, three generations, conserved charges